FULL STACK DEVELOPMENT PROJECT

SMART HOSTEL MANAGEMENT SYSTEM

A web-based system to organize student records, room allocation and hostel services.

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1. Project Introduction

The Smart Hostel Management System is a website prototype that brings common hostel administration activities into one place.

What it manages

Student registration, room details, contact messages and registration history.

Who can use it

Hostel administrators and students who need clear hostel information.

Main benefit

A simple, organized interface that reduces manual record handling.

Project type: Frontend prototype using HTML, CSS and JavaScript

2. Aim and Objectives

AIM

To design and develop a user-friendly hostel website to manage student information and basic hostel activities efficiently.

OBJECTIVES

- Provide clear navigation between website sections.
- Register students and display their details.
- Show hostel information, facilities and services.
- Maintain a browser-based history of registered students.
- Demonstrate a basic admin login and contact form.

3. Main Website Modules

Home

Welcome message and summary cards for students and rooms.

About

Hostel introduction and facilities.

Services

Student registration, room, fee and complaint service information.

Contact

Office information and message form.

Login

Demo admin username and password validation.

Student Details & History

Register student data and view saved records.

4. System Workflow

1	**Open website**	The home page is displayed.
2	**Choose a section**	Use Home, About, Services, Contact or Login.
3	**Register student**	Enter name, roll number, course, phone and room.
4	**Validate details**	The system checks required fields and duplicate roll number.
5	**Save and view**	Student data is saved in browser local storage and shown in History.

5. Technologies Used

HTML

Creates the structure and content of the website.

CSS

Adds colours, layout, spacing, cards and hover effects.

JavaScript

Handles navigation, form validation and page interactions.

Local Storage

Stores student records in the same browser.

VS Code + Browser

Used to write, test and run the project.

6. Features and Current Scope

IMPLEMENTED FEATURES

- Working navigation between sections
- Responsive blue-and-white design
- Student registration form
- Duplicate roll-number check
- Student history using local storage
- Demo login and contact confirmation

CURRENT LIMITATIONS

- No central online database
- Login is only a demonstration
- Contact form does not send email
- Records are stored only in one browser
- Room availability is a basic prototype count

Future enhancement: connect a backend (such as Flask) and a database (such as MySQL).

7. Conclusion

The Smart Hostel Management System demonstrates how web technologies can simplify hostel-related information and student record management.

The project provides a foundation that can be expanded with secure login, a backend and a database.

THANK YOU